



Installation Instruction

Installation Instruction Number: 4907834

Released Date: 08-aug-2009

Water Pump Dust Shield (Excluder)

Water Pump Dust Shield (Excluder)

Note : This is a product improvement and is not subject to campaign.

Note : This kit is to be installed when the water pump is replaced.

Note : Use the following procedure in Service Manual, Signature™, ISX, and QSX15, Bulletin 3666239. Refer to Procedure 008-062 in Section 8.
(/qs3/pubsys2/xml/en/procedures/10/10-008-062-tr.html)

Preparatory Steps

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

- Make sure the cooling system is drained with the radiator cap removed.
- Remove the belts (water pump belt, fan drive belt, and alternator belt).
- Remove the alternator.
- Remove the water pump (old dust shield is no longer required).

Install

Install the dust shield (excluder) onto the replacement water pump; two longer capscrews will need to be used, which are provided with the kit.

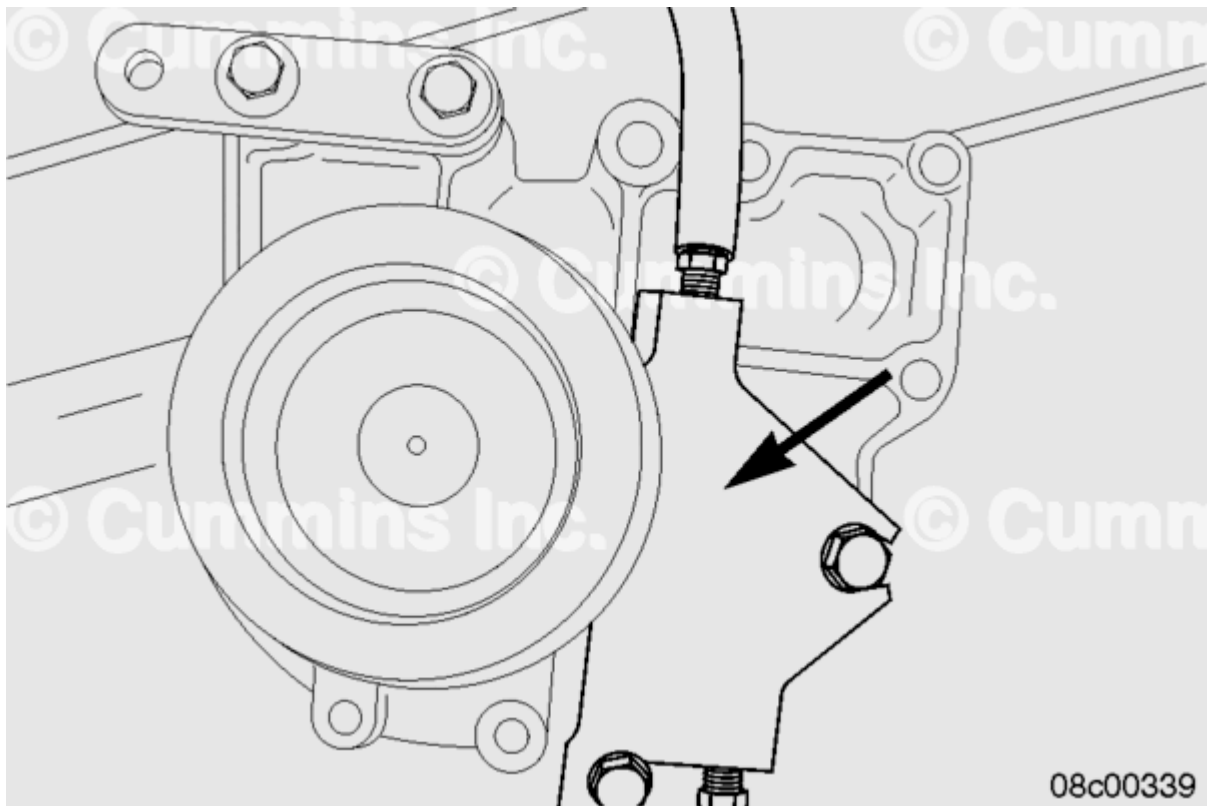
(Make sure the longer capscrews are installed in the dust shield and the water pump before installing the water pump on the engine). (Make sure the foam gasket on the dust shield is seated in place).

Install the two barbed fittings into the dust shield.

Tighten the fittings.

Torque Value: 20 N•m [177 in-lb]

Push the hoses onto the barbs of the fittings in the dust shield.

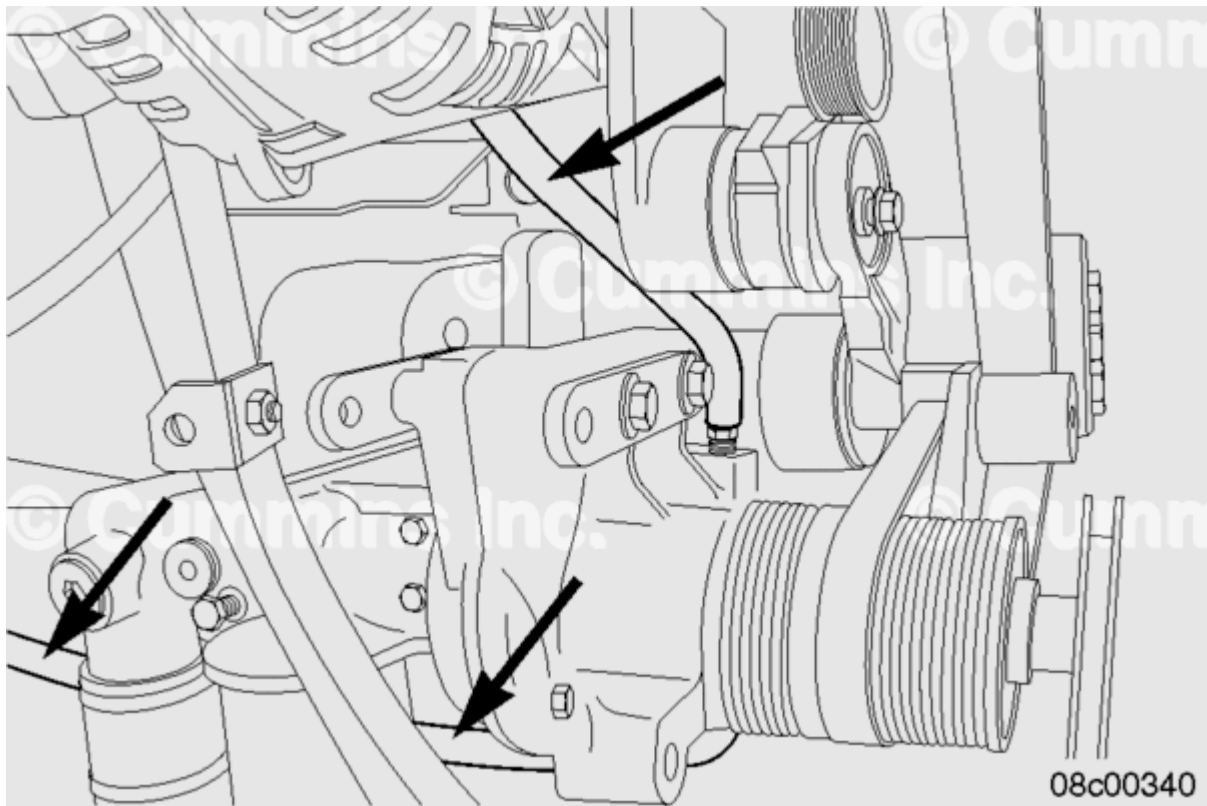


Install the water pump on the engine (use new o-rings).

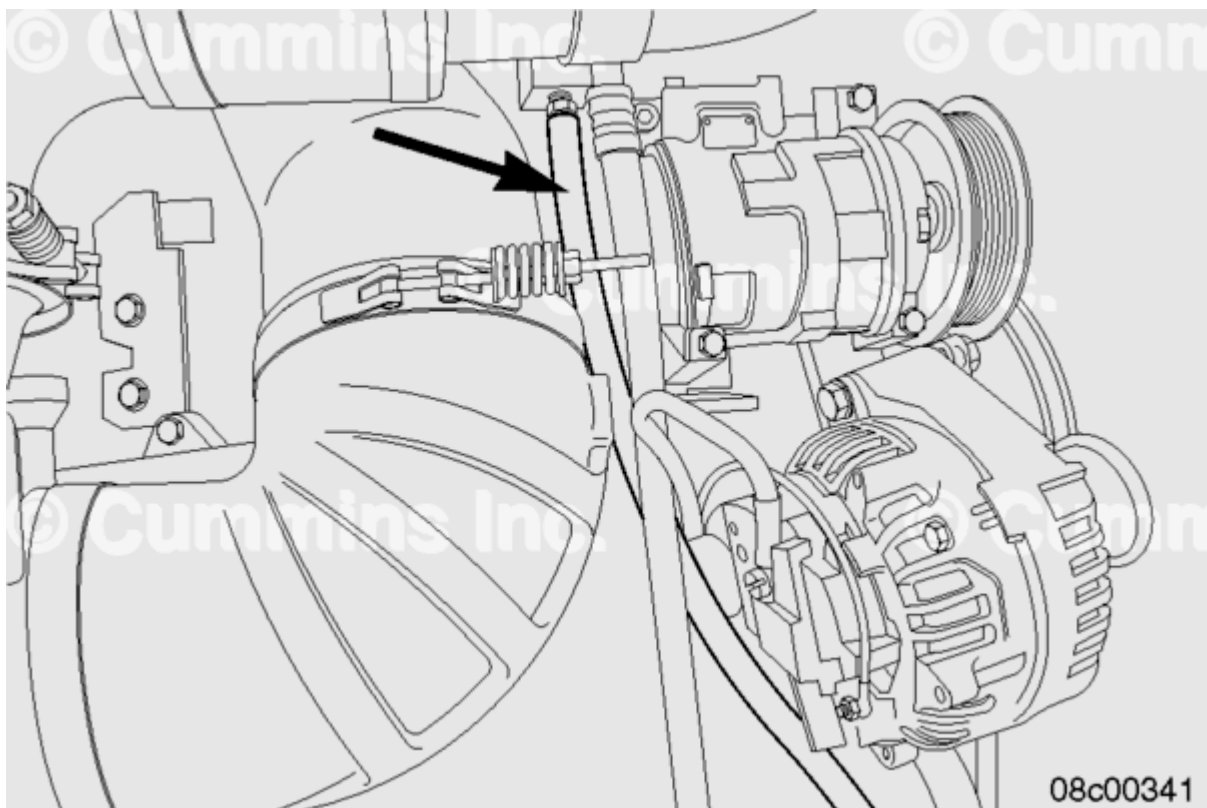
Note : The longer capscrews will need to be tightened with an open end wrench.

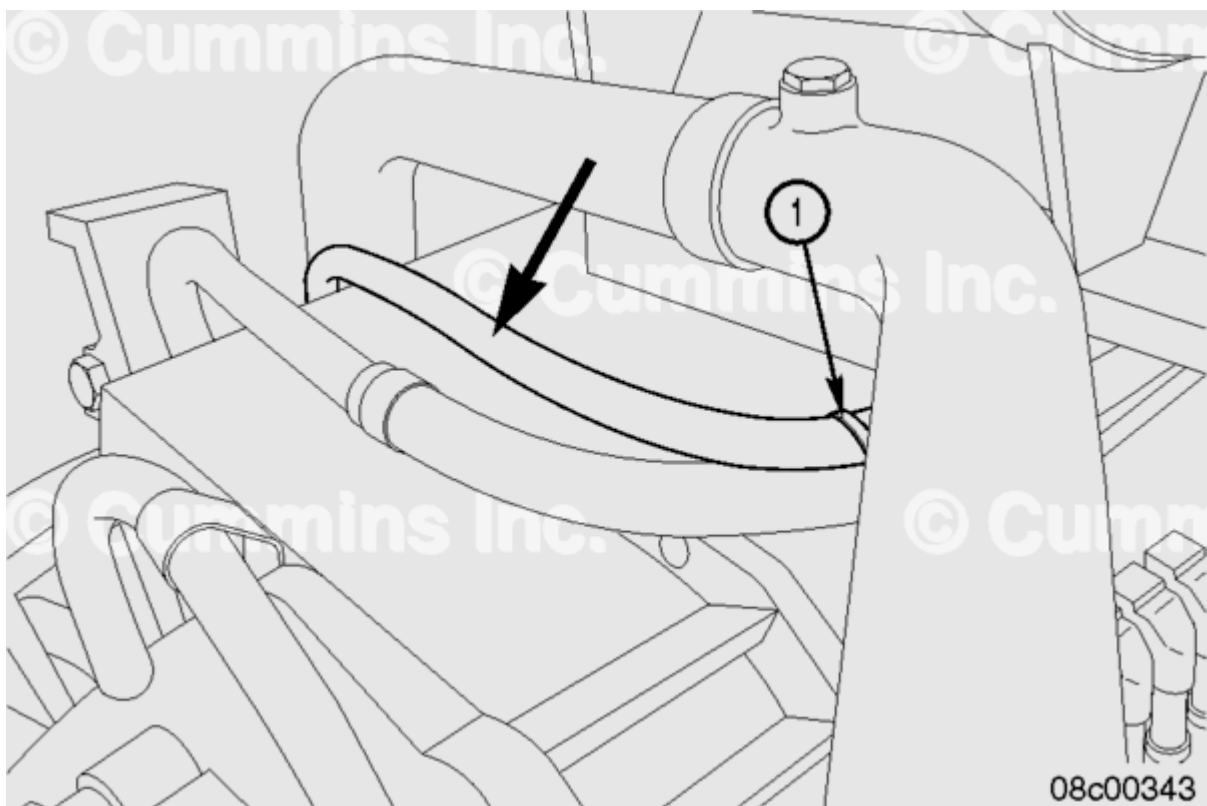
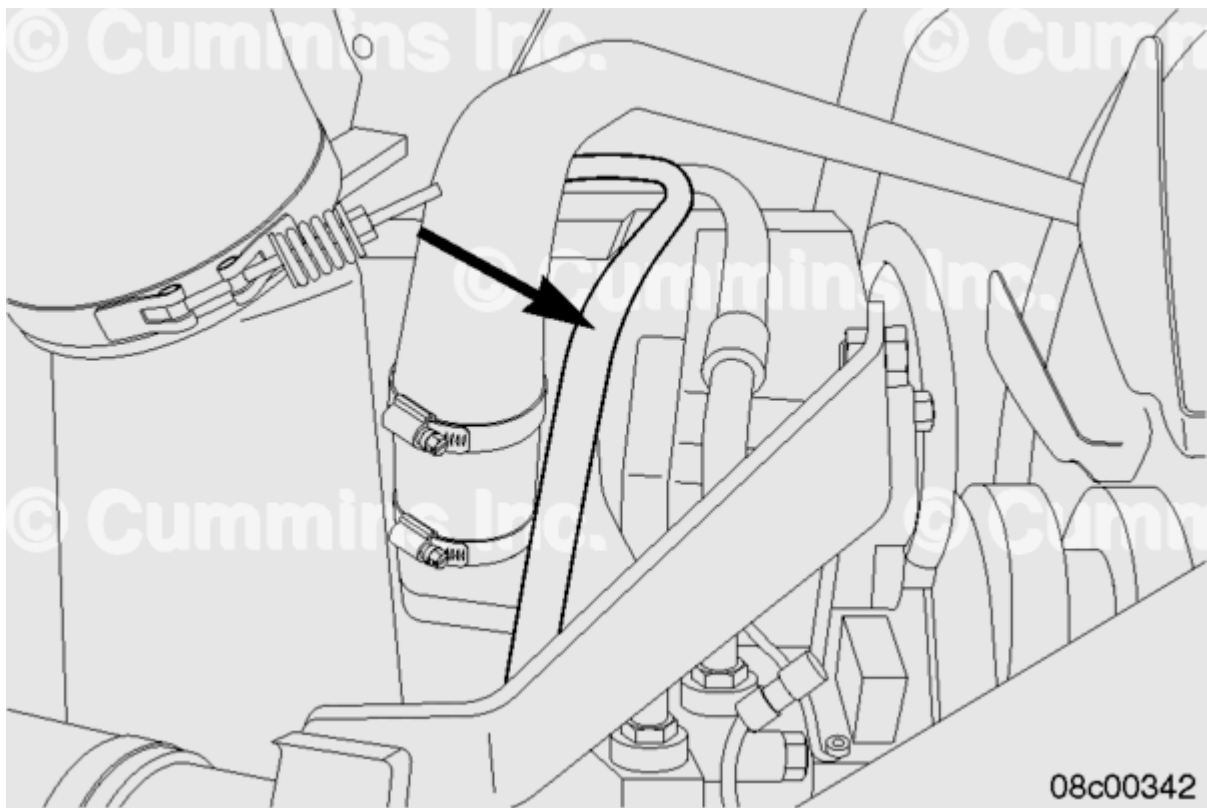
Route hoses as shown, securing them with the wire ties provided in the kit.

Note : The hoses are necessary to move the vent ports for the water pump away from the front of the engine where the vent ports are more susceptible to dirt intrusion.

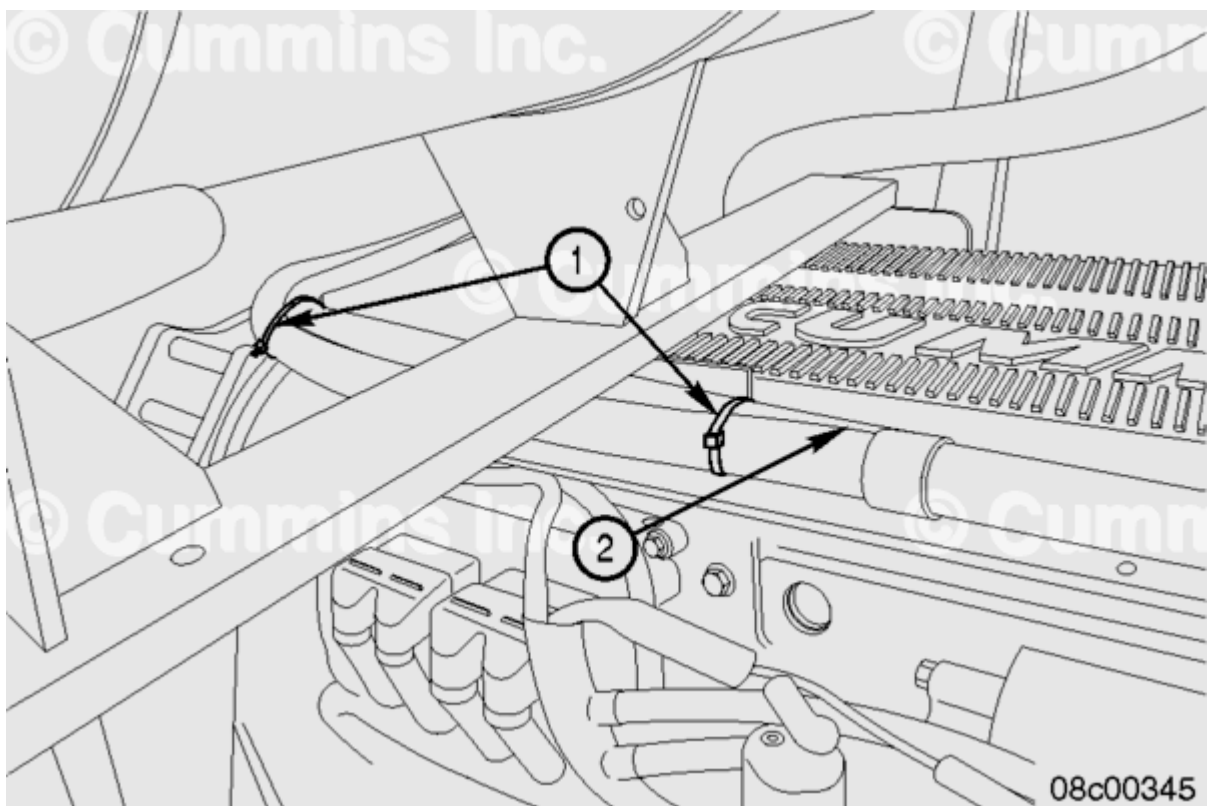
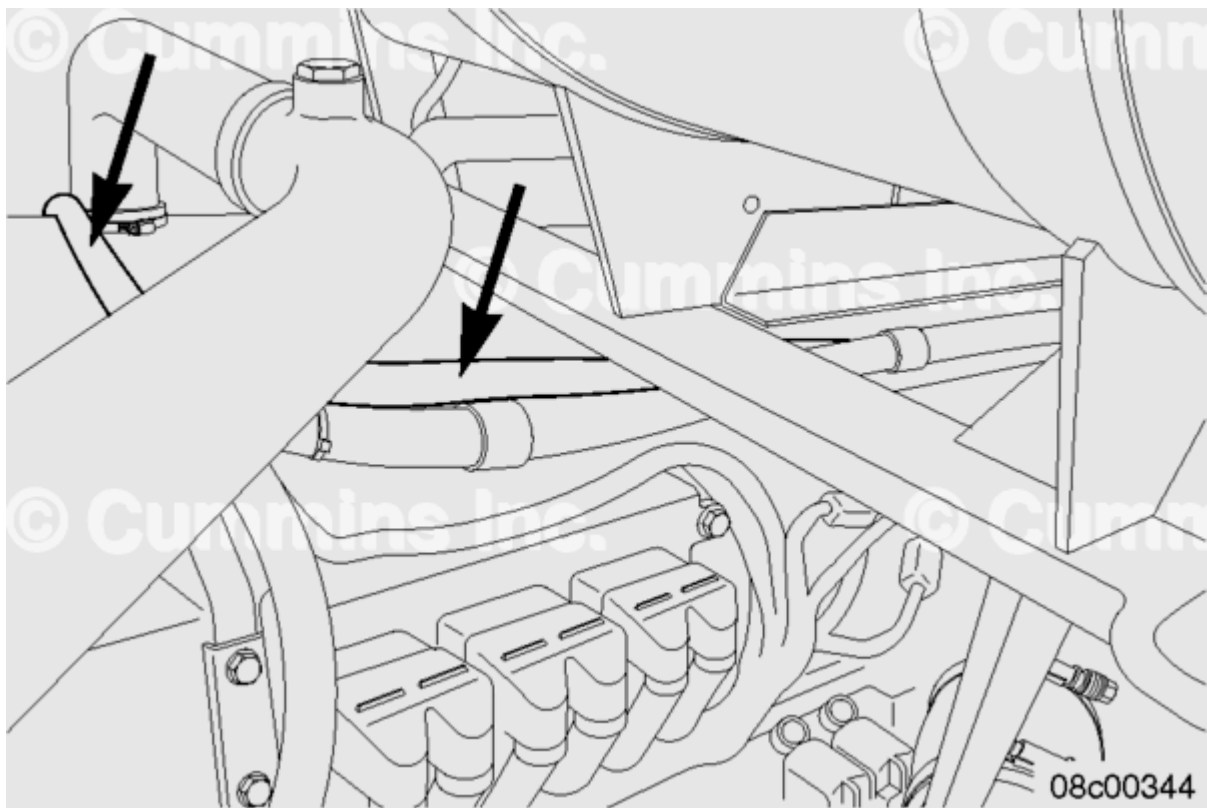


The top hose, Part Number 179972, routes up over the rocker lever cover and along the fuel rail.





1. Wire tie.

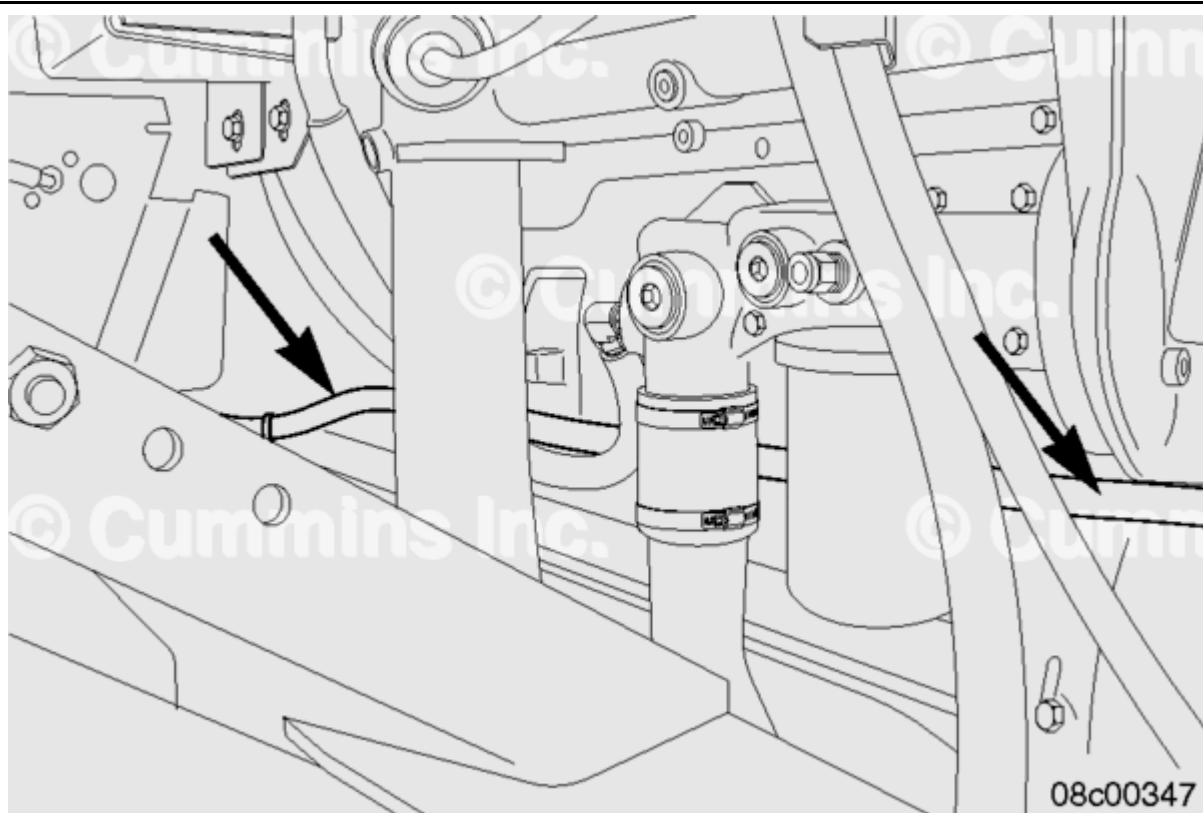
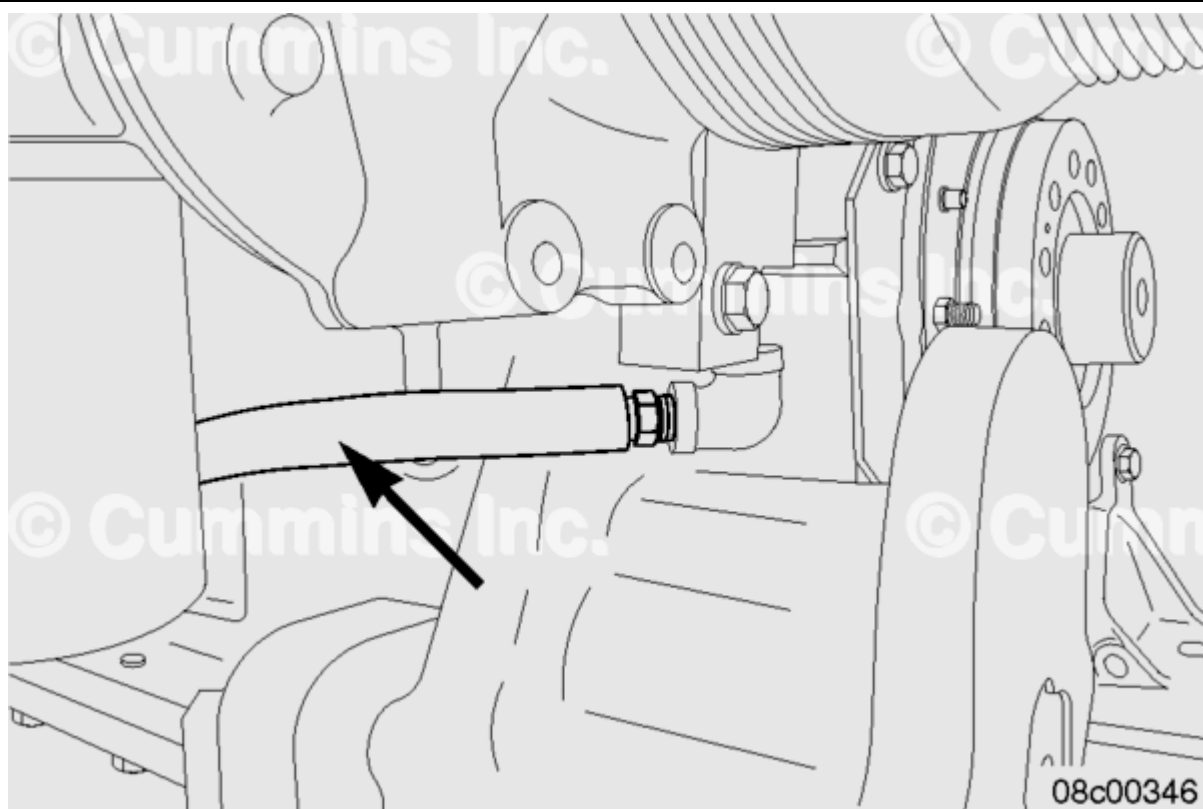


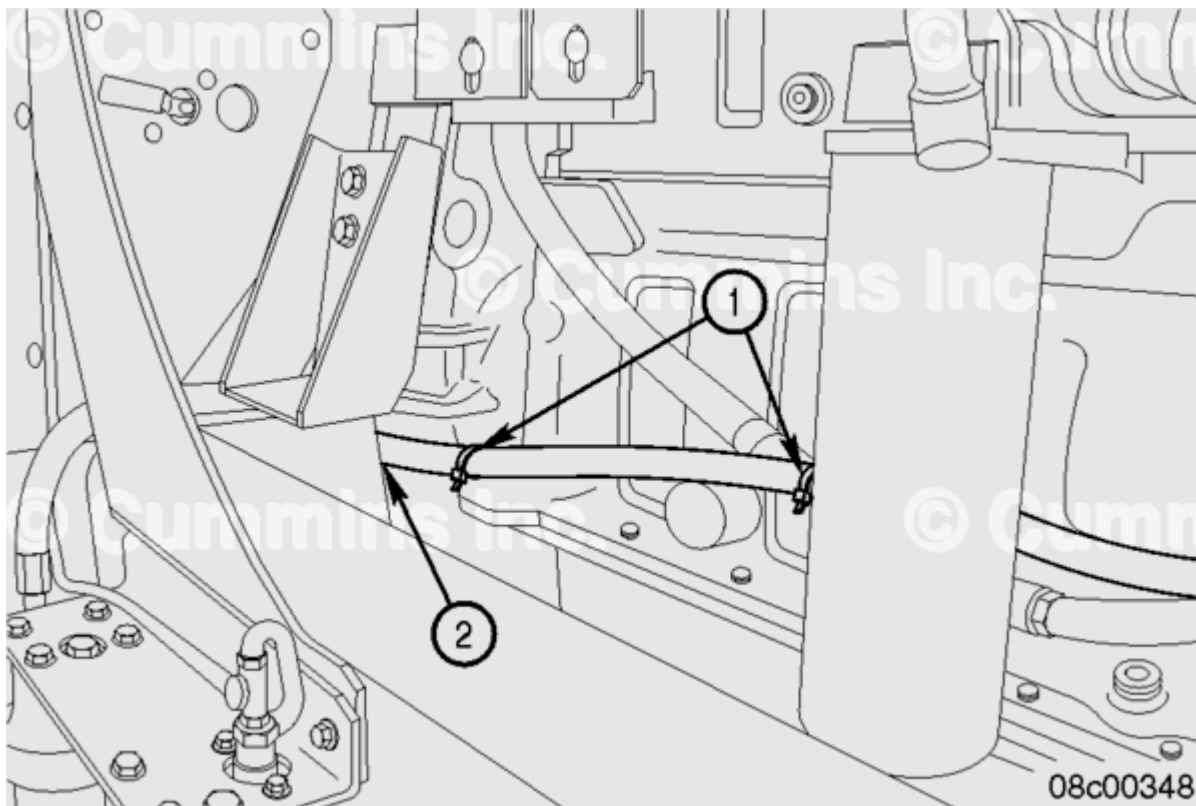
1. Wire tie

2. Termination point of top hose.

Bottom hose, Part Number 179946, is routed above the oil pan rail.

Note : The bottom hose must stay below the bottom fitting of the dust shield (make sure there is no upward slope on the hose).





1. Wire ties

2. Termination point of the bottom hose is behind the frame rail at the flywheel housing.

Finishing Steps

- Install the alternator.
- Install the belts (water pump belt, fan drive belt, and alternator belt).
- Fill the cooling system.
- Install the radiator cap.
- Make sure the coolant drain hose is closed.

Document History

Date	Details
xxx	Module Created
2012-10-29	Add part numbers to the instructions for reference as both tubes are of different lengths